LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	07
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using Hot end deterioration analysis
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using Hot end deterioration analysis
5. Specific Objectives	26.1.1.6 Trainee will be able to explain the Hot end deterioration EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of EHM trend charts; Safety: No live engine access; Supervision: Instructor monitors chart interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary components of the engine hot section?
(b) Why is monitoring hot end deterioration critical for engine life?
(c) List three factors that contribute to hot end component degradation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Hot End Deterioration • Mechanisms of Creep and Fatigue • Role of Thermal Barrier Coatings • Impact of Foreign Object Damage • Significance of Turbine Blade Cooling	L & Ds	PPT, IB	15 min
(b)	• Principles of EHM Data Acquisition • Analysis of Trend Charts • Interpretation of Deterioration Rates • Thresholds for Maintenance Action • Comparison of Baseline vs Current Data	L & Ds	PPT, PDF	20 min
(c)	• Predictive Maintenance Strategies • Cost-Benefit of EHM Implementation • Integration with Engine Overhaul Cycles • Documentation of EHM Findings • Reporting Deterioration to Maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between creep rate and hot end deterioration.
(b) How does EHM data influence the decision for an engine overhaul?

12. Home Assignment and Follow Up	Prepare a brief summary of the key differences between hot end and cold end deterioration mechanisms to be discussed in LP 08.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using vibration analysis	Refer to the master syllabus reference list	LP 08

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min